

Thomas Marconi

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EDUCATION

Georgia Institute of Technology

M.S. Computer Science | Part-Time

Florida State University

B.S. Computer Science

Atlanta, GA (Remote)

Expected Dec 2027

Tallahassee, FL

May 2023

EXPERIENCE

Tactical Air Support

Software Engineer 2

Software Engineer 1

Jacksonville, FL

Feb 2025 – Present

Feb 2024 – Feb 2025

- Led a 3-person team in building a real-time collaborative task platform using TypeScript/Node.js with React, Socket.io, PostgreSQL, and a custom RBAC system.
- Replaced 3 enterprise vendor systems with full-stack solutions (Next.js, Node.js, FastAPI, PostgreSQL), resulting in \$850K in annual licensing savings.
- Enhanced security across 8+ microservices by eliminating static credentials and implementing runtime secrets injection via Managed Identity and Key Vault.
- Launched a RAG pipeline using LangGraph and Azure Semantic Search, reducing query time from 10 minutes to seconds and cutting QA deadline by 3 weeks.
- Unified Smartsheet and QT9 data sources into a FastAPI/MongoDB backend, creating a live operational dashboard and enabling 60% faster compliance reporting.
- Containerized 6+ applications using Docker and Podman, cutting deployment time by 30+ minutes and reducing bare-metal redeployment from hours to minutes across on-premises infrastructure.
- Implemented continuous integration and continuous delivery (CI/CD) pipelines using GitLab CE, accelerating the software release cycle by 35% and reduced manual errors.
- Spearheaded the implementation of automated backend testing frameworks using Jest, leading to a 40% reduction in bug reports post-deployment and improved code reliability.
- Steered Agile transformation by conducting daily stand-ups, bi-weekly sprints, planning, retrospectives, and stakeholder meetings, resulting in a 50% boost in team velocity and predictability.

PERSONAL PROJECTS

Project Desert | *Next.js, Python (FastAPI), PostgreSQL, Docker*

Jan 2025 – Present

- Enforced explicit API contracts via OpenAPI schema validation and auto-generated TypeScript clients, reducing integration failures by 50% across all API services.
- Integrated stateless JWT authentication with server-side refresh token rotation, securing 27 API endpoints from unauthorized requests.

CityScale | *React, FastAPI, PCA* | 3rd Place – AI for Good Hackathon

Oct 2025

- Spearheaded the design of housing risk indices across 25 U.S. cities using PCA, trained across 4,519 census tracts, achieving 66% explained variance across 6 principal components.
- Engineered a full stack geospatial affordability platform featuring a real-time, user-weighted PCA scoring engine and an LLM-powered chatbot to analyze 587,754 data points across 176 indicators, achieving top 3 ranking.

TECHNICAL SKILLS

Languages: TypeScript, Python, SQL, JavaScript, C++, Java, Bash

Backend: Node.js, FastAPI, WebSockets, REST APIs, PostgreSQL, Docker, Podman

AI & Data: Vector Retrieval (ChromaDB, Azure Semantic Search), LLM Orchestration (LangGraph), Pandas, scikit-learn

Cloud & Infrastructure: Azure (Functions, Managed Identity, Key Vault, Entra ID), CI/CD, Linux

Frontend: React, Next.js, Zustand, TanStack Query, Tailwind CSS

CERTIFICATIONS

Certified Scrum Master

Scrum Alliance

Jan 2025 – Present